KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · MATH

The Houston Apartment Build

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Fractions on the Building Site

Amara is helping build an apartment building in Houston. Before the lesson, warm up with these problems. Show your work.

1 Diego's tile floor is a rectangle split into 4 equal tiles, and 2 of them are blue. Draw it. Then write two different fractions that name the blue part.

2 Which fraction is equal to 1/2?

2/3 3/6 2/5 1/4

3 Fill in the missing number: 1/3 = ?/6

4 Match each word to its meaning. Write the letter next to each word.

___ **numerator**	**A.** the bottom number of a fraction
___ **denominator**	**B.** fractions that name the same amount
___ **equivalent fractions**	**C.** the top number of a fraction

5 Review (Week 11): List all the factors of 12 and all the factors of 18. Circle the greatest common factor.

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · MATH

The Houston Apartment Build

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Inspection Report

Use equivalent fractions, simplest form, and mixed numbers to finish Amara's report. Show your work.

1. Diego tiles a hallway with 30 tiles, and 12 of them are blue. What fraction of the tiles are blue? Write it in simplest form and name the GCF you divided by.

2. Write three fractions that are equivalent to 2/3. Show the n/n you multiplied by each time.

3. Which mixed number equals 23/6?

   3 5/6 3 1/6 5 3/6 6 3/5

4. Darnell needs a piece of rebar that is 4 5/8 feet long. Write the length as an improper fraction and show how you got it.

5. Writing: Amara's cousin says, "9/12 is in simplest form because 9 is odd and 12 is even." In two or three complete sentences, explain whether he is right. Use the words greatest common factor and simplest form.

★ **Challenge:** Find a fraction that is equivalent to 3/4, has a denominator between 30 and 40, and has a numerator that is a multiple of 9. Explain how you found it.

Looking Ahead: Next week (Week 13) we travel to a mountain village in Peru, where we will multiply fractions by whole numbers and by other fractions.

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · READING

Building Up Houston: Informative Writing, Paragraphs and Transitions

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Building a Paragraph

Before the lesson, show what you already know about how paragraphs are built. Do your best. It is fine to guess!

1. Which sentence would make the best TOPIC sentence for a paragraph about building on clay soil?

   Clay soil makes building a foundation tricky. | The crew eats lunch at noon. | My favorite color is brown.

2. Circle the transition word, then write its job (order, example, contrast, or effect): The rain lasted all day. As a result, the crew stayed home.

3. Cross out the sentence that does NOT belong: Cranes lift heavy loads. They can lift steel beams to the top floor. I like tacos. Some cranes are taller than the building.

4. Add the missing comma: Next the crew pours the concrete.

5. Review (Week 11): Which sentence is punctuated correctly?

   "Watch the crane," Andre said. | "Watch the crane." Andre said. | "Watch the crane", Andre said.

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · READING

Building Up Houston: Informative Writing, Paragraphs and Transitions

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Paragraph Builder

Now use everything you practiced today: topic sentences, supporting details, transition words, and concluding sentences.

1. Write the letters in the best order to make a paragraph: A. Next, they tie steel bars into grids. B. Building the frame takes teamwork. C. Finally, concrete is poured around the steel. D. First, workers set up the forms.

2. Fill in each blank with the best transition (however, for example, as a result): Houston gets strong storms. _____, builders install thick windows. Many workers help. _____, electricians run wires. Concrete cracks when stretched. _____, steel is strong when stretched.

3. Find the off-topic sentence and write a better supporting detail to replace it: Inspectors keep buildings safe. They check the foundation. My uncle has a red truck. They test the wiring.

KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · READING

Building Up Houston: Informative Writing, Paragraphs and Transitions

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Post-Lesson Challenge: Paragraph Builder

4 Rewrite with correct capitals and commas: first the crew in houston pours concrete. however the rain slows them down.

5 Writing: Write an informative paragraph (5 to 6 sentences) about a building, bridge, barn or other structure in your community. Include a topic sentence, at least three supporting details, at least two transition words, and a concluding sentence. Underline your topic sentence and circle your transitions.

★ **Challenge:** Write a two-paragraph informative piece about how something in your community is built or made (a house, a garden, a road, a quilt, or a meal). Give each paragraph its own topic sentence and main idea, use at least four different transition words, and end the second paragraph with a concluding sentence that wraps up both.

Looking Ahead: Week 13: Figurative language and poetry. You will explore similes, metaphors and personification in original poems set in a mountain village in Peru.

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 12 · SCIENCE

Energy and Electricity

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. What is energy? Name three things at home that use energy.

2. Draw a battery, a wire and a bulb connected so you think the bulb would light. Use arrows to show where you think electricity goes.

3. Predict: Which do you think lets electricity pass through: a metal spoon, a plastic spoon, or both? Why?

4. Where do you think the electricity in your home comes from before it reaches the wall?

5. What is one thing you wonder about electricity?

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 12 · SCIENCE

Energy and Electricity

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Trace one energy chain from today (for example, a flashlight or a soccer kick). Name each form of energy, in order.

2. Explain what a circuit needs for a bulb to light, and what a switch does.

3. Use the circuit tester data: name two conductors and two insulators. Then explain the surprise about salt water or graphite.

4. Choose Lewis Latimer or Granville T. Woods. What problem did he solve, and how did his work help people?

★ **Challenge:** Design a 'safe cord' for a toy robot. Draw it and label which parts must be conductors and which must be insulators, and explain why for each part.

Looking Ahead: Next week we study forces and motion: pushes, pulls, friction, gravity and simple machines. Question to bring: Why does a ball rolling on grass stop sooner than one rolling on a smooth floor?

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 12 · SOCIAL STUDIES

Mexico, the Southwest and the Gold Rush

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. What do you already know about Mexico, Texas or California in the 1800s? Write two things.

2. Have you heard of the Alamo or the Gold Rush? Write what you know about one of them.

3. Prediction: Imagine a new border is drawn and your town is suddenly part of another country. You did not move. What might change for your family? Write or draw your ideas.

4. Write one question you have about the people who lived in the Southwest before 1848.

KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 12 · SOCIAL STUDIES

Mexico, the Southwest and the Gold Rush

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

What Did I Learn?

Use evidence from today's lesson. Write in complete sentences.

1. Explain the saying "We didn't cross the border; the border crossed us" in your own words.

2. Name one group that gained from the Gold Rush and one group that paid a high cost. Give a reason for each.

3. Was the Foreign Miners' Tax fair? Use the word "discrimination" and one fact from the lesson.

4. Using the treaty and the What Happened Next chart, explain one promise the treaty made and whether it was kept.

★ **Challenge:** Write a short letter from Juan Seguín, Biddy Mason or Norman Asing to the leaders of the 1850s. Explain what was promised, what really happened, and what you want them to do.

Looking Ahead: Next week we study slavery, abolition and the Civil War. Bring this question: when the United States gained so much new land, why did people fight about whether slavery could spread into it?